

100MHz bandwidths. These devices offer advanced visualisation, including a 256-level intensity grading and colour temperature display, and can deliver up to 500,000 waveforms per second. They include functionalities like history recording, mask test, and bode plot analysis, and are designed with a 25.6cm touchscreen and remote control capabilities.

SIGLENT
<https://siglentina.com>

Microwave signal generator with advanced analogue capabilities

Rohde & Schwarz has launched the R&S SMB100B microwave signal generator, designed for high-end applications like radar testing and semicon-



ductor evaluation. Capable of generating signals up to 40GHz, it features low phase noise, high non-harmonic suppression, and minimal wideband noise. The generator includes a high-performance oscillator for better phase noise and frequency stability. It offers up to 25dBm output power at 20GHz and 19.5dBm at 40GHz, with adjustments made through a keycode. Additional features include a compact design fit for a 48.26cm (19-inch) rack, high output level accuracy, user correction for frequency adjustments, and closed-loop power control. It also supports customised menus, automated measurements, SCPI macro recording, and emulation of legacy instruments for easy integration.

Rohde & Schwarz
<https://www.rohde-schwarz.com>

A comprehensive digital microscope

The Andonstar AD210 is a digital microscope with a 25.6cm IPS display supporting 1080P video and 12MP

photos. It offers up to 260x magnification and a 25.9cm working distance on a durable metal stand, suitable for inspecting items like large coins. Optimised for tasks such as electronics repair and biological observation, it features a bottom light stage, adjustable stand up to 30cm, and accessories for improved visibility. It includes a 32GB memory card, remote control, and camera functionality for easy image and video capture and sharing.

Andonstar Technology Co., Ltd
<https://andonstar.com>

MISCELLANEOUS

13MP camera kit for AI platforms

THine Solutions, Inc. has released THEIA-CAM 13MP PDAF Camera, THSCJ101, tailored for NVIDIA Jetson Orin NX and Nano platforms. This camera kit, which includes ISP firmware and a Linux driver, enables Jetson users to enhance their systems with advanced imaging capabilities. Designed to cater to a wide range of users, from developers in robotics and AI to educational and research institutions, the kit supports diverse applications in industrial automation, security, and surveillance. It features phase detection autofocus (PDAF), high-resolution options, and guarantees consistent image quality for mass production.

THine Solutions, Inc.
<https://www.thinesolutions.com>

Enhancing road safety with tire pressure monitoring

TREEL Mobility Solutions offers a tire pressure monitoring solution (TPMS) to improve road safety and vehicle efficiency, compatible with a wide array of vehicles from two-wheelers to heavy commercial vehicles and

off-road applications. The TPMS has three models: the first includes tire-specific sensors monitoring pressure and temperature using the Nordic Semiconductor nRF52810 SoC; the second model integrates these sensors



with Bluetooth connectivity, facilitating data transmission to the vehicle's controller area network or a mobile app; the third variant adds a dedicated receiver displaying tire status on a screen. Additionally, the Smart Tire app enhances user interaction by allowing customisation of alerts and settings to optimise battery life, supported by the energy-efficient nRF52810 SoC.

TREEL Mobility Solutions
<https://www.indiamart.com>

FPGA prototyping systems for advanced SoC design

Cadence Design Systems, Inc. has introduced the advanced Palladium Z3 Emulation and Protium X3 FPGA Prototyping systems, significantly upgrading from previous Z2 and X2 versions to address the complexities of modern SoC design.

The Palladium Z3

includes a new emulation processor to enhance compile speeds and debugging capabilities, while the Protium X3 accelerates pre-silicon software validation for extensive designs. These systems offer unified compilers and interfaces for seamless emulation-to-prototyping workflows, doubling capacity and increasing performance by 1.5 times. Integrated with NVIDIA's BlueField DPU and Quantum InfiniBand, they also feature novel applications for emulation, modelling, and power analysis, supporting crucial sectors like AI and automotive with robust debugging and validation tools.

Cadence Design Systems
<https://www.cadence.com>